Band-1 Hadamard Order 9124 – Classical Paley-II (q=4561); IN SPK via Paley-II; Gap 2052 Still OPEN

H_9124 Paley-II PASS; Paley-I SKIP (q=9123=3*3041 NOT prime); Kronecker FAIL (4*2281); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-21

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-21_Band1_Hadamard_9092_PaleyI`; `2026-09-21_Band1_Hadamard_8884_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_9124_paleyII/`.
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Honesty stamp (non-negotiable).
This memo claims **finite construction of order 9124 only (verified max|HHˆT-nI|=0), via classical Paley-II q=4561 (prime ===1 mod 4). 9124 IN SPK via Paley-II.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 9124: Paley-II **PASS**s ($q = 4561$ prime $\equiv 1 \pmod 4$); Paley-I **FAIL** ($q = 9123 = 3 \cdot 3041$ NOT prime); Kronecker **FAIL**s (4×2281; 2281 prime $\equiv 1 \pmod 4$ not Hadamard-admissible). **9124** folds into SPK **only** via classical Paley-II. It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **9092+2220+8884+2236+2160+2140+8564 / 1064+2128 / ...** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

Abstract

Finite construction of order **9124**.

- H_{9124} = Paley-II ($q = 4561$) – classical SPK
- Paley-I for 9124: **SKIP** ($q = 9123 = 3 \cdot 3041$ NOT prime)
- Kronecker for 9124: **SKIP** ($4 \cdot 2281$ not admissible)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-II.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_9124.py`

Results. overall_pass=True; units 90/90; wall 619.6988s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 9124 = 2(q + 1)$ with $q = 4561$ prime $\equiv 1 \pmod 4$ -> Paley-II eligible.
- Paley-I: $q = 9123 = 3 \cdot 3041$ NOT prime -> SKIP.
- Kronecker: $9124 = 4 \cdot 2281$; 2281 prime $\equiv 1 \pmod 4$ not Hadamard-admissible -> FAIL.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_9124 Paley-II	Paley-II q=4561	0	612.1499
H_9124 Paley-I	SKIP q=9123=3*3041 NOT prime	–	–
H_9124 Kronecker	SKIP 4*2281 not admissible	–	–

SPK closure note

Seeds include prior SPK through **9092** lineage (**9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; **1064+2128**; ...; **NOT** 8948/8644/8516/...) + verified **9124**. Next constructive gap **2052** (OPEN).

Quantity	Value
9124 in SPK	**True** (Paley-II q=4561)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Quantity	Value

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=2276->n=9104 Paley-I q=9103 prime ===3 mod4 – alternate SPK OR q=2282->n=9128 Paley-I q=9127 prime ===3 mod4 – alternate SPK OR q=2293->n=9172 triage – Paley-I q=9171 composite; Paley-II q=4585=5 7131 NOT prime power OR n=2052 standard SPK constructive gap – method-attack / lit OR method-attack SPK gap 2052 OR Track B / Object 1 meter await